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Fuel cell system and control method for fuel cell system

Abstract

The fuel cell system includes a fuel cell including a cell stack configured to have a reforming catalyst for generating hydrogen from hydrocarbon, a first flow path configured to supply a fuel containing hydrocarbon to the cell stack, and a second flow path configured to supply an oxidant gas to the cell stack such that the oxidant gas flows oppositely or orthogonally to the fuel. The control method for the fuel cell system including: detecting a temperature of a discharged oxidant gas that is the oxidant gas discharged from the second flow path; and performing a temperature control of the fuel cell based on the temperature of the discharged oxidant gas.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a fuel cell system and a control method for the fuel cell system, which includes a fuel cell.

BACKGROUND ART

(2) JP 2017-117550 A discloses a method of controlling a temperature distribution throughout a fuel cell to be uniform by detecting a temperature of a central portion of a cell stack where the temperature is highest. In the fuel cell system, the temperature distribution throughout the fuel cell is made uniform by supplying an inert gas to the central portion of the cell stack where the temperature is highest, in order to decrease the temperature of the central portion of the cell stack.

SUMMARY OF INVENTION

(3) As described above, JP 2017-117550 A is based on a premise that the central portion of the cell stack has the highest temperature, but depending on a configuration and/or an operation scene of the fuel cell, the central portion of the cell stack may not always have the highest temperature as compared with other portions. For example, in a fuel cell including a catalyst configured to reform a fuel (hereinafter, referred to as a reforming catalyst) inside a cell stack, when partial oxidation reforming is performed, an inlet end of the anode is relatively hotter than the central portion of the

cell stack. Therefore, in the internal reforming fuel cell, when a temperature control is performed based on a temperature of the central portion of the cell stack, a temperature may exceed an upper limit of heat resistance in a part of the cell stack.

(4) An object of the present invention is to provide a fuel cell system and a control method configured to perform a temperature control within a range not exceeding an upper limit of heat resistance in a fuel cell when a cell stack with a reforming catalyst is included.

(5) One aspect of the present invention is a control method for a fuel cell system. The fuel cell system includes a fuel cell including a cell stack configured to have a reforming catalyst for generating hydrogen from hydrocarbon, a first flow path configured to supply a fuel containing hydrocarbon to the cell stack, and a second flow path configured to supply an oxidant gas to the cell stack such that the oxidant gas flows oppositely or orthogonally to the fuel. The control method for the fuel cell system including: detecting a temperature of a discharged oxidant gas that is the oxidant gas discharged from the second flow path; and performing a temperature control of the fuel cell based on the temperature of the discharged oxidant gas.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram showing a configuration of a fuel cell system according to a first embodiment.

(2) FIG. 2 is an explanatory diagram showing a flow path of a fuel and a flow path of an oxidant gas in a cell stack.

(3) FIG. 3 is a schematic graph showing a temperature distribution in a fuel cell when partial oxidation reforming is performed.

(4) FIG. 4 is a flowchart for a temperature control of the fuel cell.

(5) FIG. 5 is a graph schematically showing a temperature control in a warm-up operation.

(6) FIG. 6 is a graph schematically showing a temperature control in a steady operation.

(7) FIG. 7 is a block diagram showing a configuration of a fuel cell system according to a second embodiment.

(8) FIG. 8 is an explanatory diagram of a modification in which an attachment position of a temperature sensor is changed.

(9) FIG. 9 is an explanatory diagram of a modification in which a flow path configuration of the fuel and the oxidant gas is changed.

(10) FIG. 10 is an explanatory diagram of a modification in which the flow path configuration of the fuel and the oxidant gas and the attachment position of the temperature sensor are changed.

DESCRIPTION OF EMBODIMENTS

First Embodiment

(11) FIG. 1 is a block diagram showing a configuration of a fuel cell system 10 according to a first embodiment. As shown in FIG. 1, the fuel cell system 10 includes a power generation device 11 and a controller 12. The power generation device 11 is a device that generates a power using a fuel cell 20, and includes, in addition to the fuel cell 20, a fuel flow rate adjustment device 31, an air flow rate adjustment device 33, an oxidant gas supply device 41, a heat exchanger 43, a combustor 51, a temperature sensor 60, and the like. The controller 12 is a control device that integrally controls the power generation device 11 and units constituting the power generation device 11.

(12) The fuel cell 20 includes a fuel cell (hereinafter, simply referred to as a cell) configured to generate a power by an electrochemical reaction using a fuel and an oxidant gas. In the present embodiment, the fuel cell 20 includes a cell stack 21 which is formed by a single cell or by stacked multiple cells and configured to generate a required power as a whole.

(13) The cell stack 21 includes a reforming catalyst inside thereof (hereinafter, referred to as an

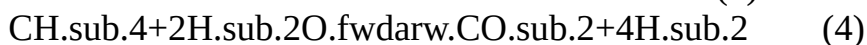
internal reforming catalyst). The internal reforming catalyst **22** reforms a fuel supplied to the fuel cell **20** (hereinafter, referred to as a raw fuel) by generating or promoting a specific chemical reaction, and generates a fuel (hereinafter, referred to as a reformed fuel (an anode gas)) that can be directly used for power generation from the raw fuel. The raw fuel is, for example, a fuel containing hydrocarbon group, such as an alkyl group, as its main component, and is specifically a hydrocarbon such as methane or an alcohol such as methanol. The internal reforming catalyst **22** generates the reformed fuel containing hydrogen used for the power generation from the raw fuel containing hydrocarbons. Therefore, the fuel cell **20** is a so-called direct internal reforming type, and in principle, the fuel cell **20** can generate a power by itself by supplying an unreformed raw fuel or an incompletely reformed raw fuel without requiring a separate reformer.

(14) Fuel reforming that can be performed by the cell stack **21** includes, for example, partial oxidation reforming (PDX), steam reforming (SR), and carbon dioxide reforming. The internal reforming catalyst **22** contributes to a part or all of these reforming reactions.

(15) Partial oxidation reforming is a reforming reaction in which the raw fuel is partially oxidized by mixing oxygen (O.sub.2) with the raw fuel in order to obtain the reformed fuel. For example, when the raw fuel is methane (CH.sub.4), partial oxidation reforming proceeds mainly by partial oxidation reaction shown in the following (1). In partial oxidation reforming, for example, complete combustion (complete oxidation) shown in the following (2) occurs and proceeds in an initial stage, except for a case where partial oxidation reforming is occurred on some very exceptional catalysts. Therefore, in partial oxidation reforming, water (H.sub.2O) is also generated in addition to hydrogen (H.sub.2). Both partial oxidation reaction of (1) and complete combustion of (2) are exothermic reactions. Therefore, partial oxidation reforming is an exothermic reaction. As can be seen from (1) and (2), partial oxidation reforming is a reaction that consumes oxygen. Therefore, partial oxidation reforming does not occur uniformly throughout the cell stack **21**, but occurs relatively more in an upstream portion of a fuel flow path (a first flow path **23** to be described later) and relatively less in a downstream portion of the fuel flow path where oxygen has been consumed. This is because oxygen is abundant in the upstream portion, but is consumed sequentially by the progress of partial oxidation reforming, and contained oxygen is relatively low in the downstream portion. In this way, since the occurrence of partial oxidation reforming varies gradually along the fuel flow path in the cell stack **21**, a large temperature gradient may be generated in the cell stack **21**.



(16) Steam reforming is a reforming reaction in which the raw fuel is partially oxidized by reacting the raw fuel or the reformed fuel with water (steam) in order to obtain the reformed fuel. For example, when the raw fuel is methane, steam reforming proceeds mainly by an oxidation reaction shown in the following (3) or (4). Steam reforming may proceed using newly supplied water or water generated by partial oxidation reforming (particularly, complete combustion). In the present embodiment, the cell stack **21** performs steam reforming using water generated by partial oxidation reforming. Steam reforming is an endothermic reaction.



(17) Carbon dioxide reforming is a reforming reaction in which the raw fuel or the reformed fuel is partially oxidized to obtain the reformed fuel, by reacting the raw fuel, or the reformed fuel reformed by partial oxidation reforming or steam reforming, with carbon dioxide (CO.sub.2). For example, when the raw fuel is methane, carbon dioxide reforming proceeds mainly by partial oxidation reaction shown in the following (5). Carbon dioxide reforming may proceed by using newly supplied carbon dioxide or carbon dioxide generated by partial oxidation reforming (particularly, complete combustion) or steam reforming. In the present embodiment, the cell stack **21** performs carbon dioxide reforming using carbon dioxide generated by partial oxidation

reforming and/or steam reforming. Carbon dioxide reforming is an endothermic reaction.

$\text{CH.sub.4} + \text{CO.sub.2} \rightarrow \text{fwdarw.2CO} + 2\text{H.sub.2}$ (5)

(18) The fuel cell **20** is, for example, a solid oxide fuel cell (SOFC). When the fuel cell **20** is the SOFC, the cell stack **21** includes the first flow path **23** which is a flow path of a raw fuel and/or a reformed fuel (hereinafter, simply referred to as a fuel unless a distinction is necessary), a second flow path **24** which is a flow path of an oxidant gas, and an electrolyte (not shown) made of a solid oxide. The cell stack **21** has a structure in which the first flow path **23** and the second flow path **24** are adjacent to each other via the electrolyte. Therefore, the internal reforming catalyst **22** is provided in the first flow path **23** which is the flow path of the fuel. An electrode (hereinafter, referred to as an anode) configured to act as a negative electrode is provided on a surface of the electrolyte in the first flow path **23**, and an electrode (hereinafter, referred to as a cathode) configured to act as a positive electrode is provided on a surface of the electrolyte in the second flow path **24**. In the present embodiment, the internal reforming catalyst **22** is separate from the anode, but a part or all of the anode may act as the internal reforming catalyst **22**. The internal reforming catalyst **22** is provided substantially uniformly throughout the cell stack **21**. That is, the internal reforming catalyst **22** is provided not only in a specific portion of the first flow path **23**, and the raw fuel may be reformed substantially at any portion of the first flow path **23** in the cell stack **21**.

(19) The fuel cell **20** includes an anode inlet **A1**, an anode outlet **A2**, a cathode inlet **C3**, and a cathode outlet **C4**.

(20) The anode inlet **A1** is an end portion on an upstream side of the first flow path **23**, and is an inlet of the raw fuel to the fuel cell **20**. Therefore, the anode inlet **A1** is connected to the fuel flow rate adjustment device **31** via a fuel supply path **32**. On the other hand, the anode outlet **A2** is an end portion on a downstream side of the first flow path **23**, and discharges the used fuel, a gas generated by the power generation, and the like (hereinafter, referred to as a discharged fuel (an anode off-gas)) to an outside of the fuel cell **20**. In the present embodiment, the anode outlet **A2** is connected to the combustor **51** via a fuel discharge path **34**.

(21) The cathode inlet **C3** is an end portion on an upstream side of the second flow path **24**, and is an inlet of the oxidant gas to the fuel cell **20**. In the present embodiment, the cathode inlet **C3** is connected to the oxidant gas supply device **41** via an oxidant gas supply path **42**. On the other hand, the cathode outlet **C4** is an end portion on a downstream side of the second flow path **24**, and discharges the used oxidant gas for the power generation, a gas generated by the power generation, and the like (hereinafter, referred to as a discharged oxidant gas (a cathode off-gas)) to the outside of the fuel cell **20**. In the present embodiment, the cathode outlet **C4** is connected to the combustor **51** via an oxidant gas discharge path **44**.

(22) The fuel cell **20** supplies a power to an apparatus, a device, or the like connected to the fuel cell **20**. The apparatus, the device, or the like connected to the fuel cell **20** is, for example, a storage battery, or a motor used as a drive source of an electric vehicle. In the present embodiment, although not shown, the fuel cell **20** is connected to a storage battery connected to a motor for driving an electric vehicle. Therefore, the fuel cell **20** acts as an energy source that indirectly drives the electric vehicle.

(23) The fuel flow rate adjustment device **31** is a device (a fuel supply device) that supplies the raw fuel to the fuel cell **20**, and is a device that adjusts a flow rate of the raw fuel to be supplied to the fuel cell **20**. The fuel flow rate adjustment device **31** is connected to the anode inlet **A1** via the fuel supply path **32**. The fuel flow rate adjustment device **31** includes a fuel tank and an injector. The fuel tank stores the raw fuel, for example, in a liquid state. The injector injects the raw fuel from the fuel tank into the fuel supply path **32**. Accordingly, the fuel flow rate adjustment device **31** supplies the raw fuel to the fuel cell **20**. A timing at which the fuel flow rate adjustment device **31** supplies the raw fuel to the fuel cell **20**, a supply amount of the raw fuel, and the like are controlled by the controller **12**. The fuel flow rate adjustment device **31** may include a heat exchanger or the

like as necessary. In this case, the fuel flow rate adjustment device **31** can adjust a temperature, a phase, a pressure, and the like of the raw fuel by heating, heat exchange, or the like. In the present embodiment, the fuel flow rate adjustment device **31** vaporizes the raw fuel and supplies the vaporized raw fuel to the fuel cell **20**.

(24) The air flow rate adjustment device **33** is connected to the fuel supply path **32**, and is a device (an air supply device) that mixes the raw fuel to be supplied to the fuel cell **20** by the fuel flow rate adjustment device **31** with air as necessary, and is a device that adjusts a flow rate of the air to be mixed with the raw fuel. The air flow rate adjustment device **33** is, for example, an air blower. A purpose of mixing the raw fuel with the air by the air flow rate adjustment device **33** is to occur partial oxidation reforming in the cell stack **21** or to promote partial oxidation reforming occurred in the cell stack **21** by including oxygen in the raw fuel. Therefore, it is sufficient for the air flow rate adjustment device **33** to be able to mix the oxygen with the raw fuel as necessary. Therefore, the air flow rate adjustment device **33** may be configured to mix the raw fuel with oxygen alone, a gas other than air containing oxygen, air whose constituent components, temperature, and the like are adjusted, or the like, instead of natural air. A timing at which the air flow rate adjustment device **33** mixes the raw fuel with the air, an amount of air to be mixed with the raw fuel, and the like are controlled by the controller **12**.

(25) The oxidant gas supply device **41** is a device that supplies the oxidant gas to the fuel cell **20**, and is connected to the cathode inlet **C3** via the oxidant gas supply path **42**. In the present embodiment, the oxidant gas is air containing oxygen used for the power generation by the fuel cell **20**. That is, in the present embodiment, the oxidant gas supply device **41** is an air blower.

(26) The heat exchanger **43** is provided in the oxidant gas supply path **42**, and heats the oxidant gas by heat exchange. Therefore, the heat exchanger **43** supplies the heated oxidant gas to the fuel cell **20**. The heat exchange for heating the oxidant gas is performed with a gas discharged from the combustor **51** (hereinafter, referred to as a discharged gas).

(27) The discharged fuel is supplied from the fuel cell **20** to the combustor **51** via the fuel discharge path **34**. The discharged oxidant gas is supplied from the fuel cell **20** to the combustor **51** via the oxidant gas discharge path **44**. The combustor **51** combusts the discharged fuel and the discharged oxidant gas by, for example, catalytic combustion. The discharged gas generated as a result of the combustion is discharged from a gas discharge path **52** to an outside of the fuel cell system **10** via the heat exchanger **43**. Therefore, heat generated by the electrochemical reaction in the fuel cell **20**, the generation of the discharged gas, and the like is used for heating the oxidant gas in the oxidant gas supply path **42** by the heat exchange in the heat exchanger **43**.

(28) The temperature sensor **60** is provided in the oxidant gas discharge path **44**. The temperature sensor **60** detects a temperature of the discharged oxidant gas that is the oxidant gas discharged from the fuel cell **20**, more specifically, the oxidant gas discharged from the second flow path **24**. The temperature sensor **60** inputs the detected temperature of the discharged oxidant gas to the controller **12**.

(29) The controller **12** performs a temperature control of the fuel cell **20** based on the temperature of the discharged oxidant gas detected by the temperature sensor **60**. Particularly, in the present embodiment, the controller **12** performs the temperature control of the fuel cell **20** with heat generated by partial oxidation reforming in the cell stack **21**. A specific method by which the controller **12** performs the temperature control of the fuel cell **20** will be described in detail later. The controller **12** is a computer or a microcomputer implemented by using a processor such as a CPU and a GPU, a memory, and the like. The controller **12** may be implemented as a dedicated computer for controlling the fuel cell system **10**, or may be implemented as a part of a computer for controlling other devices, systems, or the like. For example, when the fuel cell system **10** is mounted on an electric vehicle, a vehicle controller that controls driving and the like of the electric vehicle may be made to act as the controller **12** of the fuel cell system **10**. The controller **12** may be implemented as a server type computer that controls the power generation device **11** by

communicating with the power generation device **11**.

(30) FIG. 2 is an explanatory diagram showing a flow path of a fuel and a flow path of an oxidant gas in the cell stack **21** in the present embodiment. As shown in FIG. 2, the first flow path **23** is a flow path that supplies the fuel to the cell stack **21**. The second flow path **24** is a flow path that supplies the oxidant gas to the cell stack **21** such that the oxidant gas flows oppositely to the fuel. The first flow path **23** and the second flow path **24** are parallel in shape as a flow path, but a direction in which the fuel flows through the first flow path **23** and a direction in which the oxidant gas flows through the second flow path **24** are opposite to each other. Therefore, in consideration of the flow directions of the fuel and the oxidant gas, the first flow path **23** and the second flow path **24** are reverse parallel or antiparallel, and the fuel and the oxidant gas form a so-called counter flow. As will be described later, the second flow path **24** may be a flow path that supplies the oxidant gas to the cell stack **21** such that the oxidant gas flows orthogonally to the fuel.

(31) FIG. 3 is a schematic graph showing a temperature distribution in the fuel cell **20** when partial oxidation reforming is performed. FIG. 3 shows the temperature distribution in the fuel cell **20** formed when the cell stack **21** reaches or exceeds a temperature at which partial oxidation reforming may be occurred as a whole and partial oxidation reforming occurs on the cell stack **21**. The temperature at which partial oxidation reforming may be occurred is, for example, about 300° C. or higher. A horizontal axis represents a position along the first flow path **23** and the second flow path **24**. The cells constituting the cell stack **21** are stacked in a direction of a vertical axis. A graph **101** indicated by a solid line represents a temperature of the fuel in the fuel cell **20**, and a graph **102** indicated by a broken line represents a temperature of the oxidant gas. The temperatures of the fuel and the oxidant gas in the cell stack **21** may be considered to be substantially the same as a temperature of the fuel cell **20**.

(32) When the raw fuel is supplied from the anode inlet **A1**, for example, at ordinary temperature, and the raw fuel reaches the cell stack **21**, partial oxidation reforming, which is an exothermic reaction, is occurred. As a result, as shown in the graph **101**, the temperatures of the fuel and the fuel cell **20** increase in accordance with an amount of partial oxidation reforming occurred. As described above, a relatively large amount of partial oxidation reforming is occurred on an anode inlet **A1** side where the oxygen is abundant, and a relatively small amount of partial oxidation reforming is occurred on an anode outlet **A2** side where the oxygen has been consumed. Therefore, as shown in the graph **101**, partial oxidation reforming forms a temperature distribution in the fuel cell **20** in which the temperature generally decreases from the anode inlet **A1** to the anode outlet **A2**. According to the temperature distribution, the fuel cell **20** has a highest temperature T_p at a predetermined position (the anode inlet **A1** or a position in the vicinity of the anode inlet **A1**) biased toward the anode inlet **A1** side, rather than a position of a central portion of the fuel cell **20**.

(33) As shown in the graph **102**, a temperature distribution of the oxidant gas follows the temperature distribution formed by partial oxidation reforming as described above, and is substantially the same temperature distribution at least in the cell stack **21**. This is because the fuel and the oxidant gas flow in the cell stack **21** via the electrolyte, which is a thin film, in an integrated manner on front and back sides, and thus have substantially the same temperature due to mutual heat exchange. Therefore, the discharged oxidant gas discharged from the cathode outlet **C4** is discharged at a temperature substantially equal to a maximum temperature in the temperature distribution formed by partial oxidation reforming, or at a temperature correlated with the maximum temperature in the temperature distribution formed by partial oxidation reforming. The temperature correlated with the maximum temperature refers to a temperature by which a maximum temperature of the fuel cell **20** can be calculated or estimated based on a certain relationship of the temperature with the maximum temperature of the fuel cell **20**.

(34) The temperature sensor **60** detects the temperature of the discharged oxidant gas. As a result, the temperature sensor **60** detects the maximum temperature of the fuel cell **20** or a temperature correlated with the maximum temperature of the fuel cell **20**. The controller **12** performs the

temperature control of the fuel cell **20** based on the temperature of the discharged oxidant gas detected by the temperature sensor **60**, thereby performing the temperature control of the fuel cell **20** based on the maximum temperature of the fuel cell **20**.

(35) In the present embodiment, the fuel and the oxidant gas form the counter flow, and thus the temperature of the discharged oxidant gas is substantially equal to the maximum temperature in the temperature distribution formed by partial oxidation reforming. Therefore, the temperature sensor **60** substantially detects the maximum temperature of the fuel cell **20** by detecting the temperature of the discharged oxidant gas. The controller **12** directly performs the temperature control of the fuel cell **20** based on the maximum temperature of the fuel cell **20** by using the temperature of the discharged oxidant gas detected by the temperature sensor **60**.

(36) The temperature of the fuel cell **20** decreases in a central portion of the cell stack **21** because steam reforming and/or carbon dioxide reforming, which is an endothermic reaction, proceed. The temperature of the fuel cell **20** increases from the central portion of the cell stack **21** to the anode outlet **A2** and the cathode inlet **C3** because the heated oxidant gas is supplied from the cathode inlet **C3**.

(37) Hereinafter, a method in which the controller **12** controls the temperature of the fuel cell **20** with the heat generated by partial oxidation reforming in the cell stack **21** will be described. FIG. **4** is a flowchart for the temperature control of the fuel cell **20**. That is, when the temperature control of the fuel cell **20** is performed with the heat generated by partial oxidation reforming in the cell stack **21**, the controller **12** controls the temperature of the fuel cell **20** by repeatedly performing a discharged oxidant gas temperature acquisition step **S101**, a determination step **S102**, a temperature increase limit calculation step **S103**, a target heat generation calculation step **S104**, a fuel flow rate calculation step **S105**, a mixed air flow rate calculation step **S106**, and a control execution step **S107** in this order at predetermined time intervals. The “temperature control of the fuel cell **20**” particularly refers to a control performed in the temperature increase limit calculation step **S103**, the target heat generation calculation step **S104**, the fuel flow rate calculation step **S105**, the mixed air flow rate calculation step **S106**, and the control execution step **S107**.

(38) In the discharged oxidant gas temperature acquisition step **S101**, the controller **12** acquires the temperature of the discharged oxidant gas detected by the temperature sensor **60**. That is, the controller **12** is configured to act as a discharged oxidant gas temperature acquisition unit. During a warm-up operation of the fuel cell system **10** (before the occurrence of partial oxidation reforming), a temperature on a cathode outlet **C4** side (the anode inlet **A1** side) decreases.

Therefore, during the warm-up operation, the controller **12** can determine whether a temperature of the fuel cell **20** reaches a temperature at which partial oxidation reforming is possible, based on the temperature of the discharged oxidant gas detected by the temperature sensor **60** that reflects the temperature. On the other hand, when partial oxidation reforming is occurred, the temperature of the discharged oxidant gas detected by the temperature sensor **60** reflects the heat generated by partial oxidation reforming and is a temperature substantially equal to the maximum temperature of the fuel cell **20**. Therefore, after the occurrence of partial oxidation reforming, the controller **12** regards the temperature of the discharged oxidant gas acquired from the temperature sensor **60** as the maximum temperature of the fuel cell **20**, and performs the following steps for controlling the temperature of the fuel cell **20**. The temperature of the discharged oxidant gas may be corrected as necessary, when the temperature of the discharged oxidant gas may have deviation from the maximum temperature of the fuel cell **20** due to an operating state of the fuel cell **20** or the like and it may cause an error in control. In this case, the controller **12** performs the following steps in consideration of a correlation between the temperature of the discharged oxidant gas and the maximum temperature of the fuel cell **20**. When estimating the maximum temperature of the fuel cell **20** by using the temperature of the discharged oxidant gas, for example, a map may be used that associate the temperature of the discharged oxidant gas and the maximum temperature of the fuel cell.

(39) In the determination step S102, the controller 12 determines whether the temperature of the discharged oxidant gas is equal to or higher than the temperature at which partial oxidation reforming is possible. That is, the controller 12 is configured to act as a determination unit. When the temperature of the discharged oxidant gas is lower than the temperature at which partial oxidation reforming is possible, the controller 12 continues to acquire the temperature of the discharged oxidant gas at a predetermined cycle without executing the step after the determination step S102. This is because, in a case where the temperature of the discharged oxidant gas, that is, even the maximum temperature of the fuel cell 20, does not reach the temperature at which partial oxidation reforming can be performed, partial oxidation reforming is not occurred, and thus the heat cannot be used for the temperature control of the fuel cell 20. When the temperature of the discharged oxidant gas is equal to or higher than the temperature at which partial oxidation reforming is possible, the controller 12 performs the temperature increase limit calculation step S103.

(40) In the temperature increase limit calculation step S103, the controller 12 calculates a temperature increase limit ΔT_m using a detected temperature T_q of the discharged oxidant gas. That is, the controller 12 is configured to act as a temperature increase limit calculation unit. The temperature increase limit ΔT_m is a temperature obtained by subtracting the detected temperature T_q of the discharged oxidant gas from a heatproof limit temperature of a component having a lowest heatproof (hereinafter, simply referred to as a heatproof limit temperature) among the cell stack 21 and other components constituting the fuel cell 20. In the present embodiment, a heatproof limit temperature T_k of the internal reforming catalyst 22 is the lowest among heatproof limit temperatures of the cell stack 21 and the other components of the fuel cell 20. Therefore, the controller 12 calculates the temperature increase limit ΔT_m by $\Delta T_m = T_k - T_q$.

(41) In the target heat generation calculation step S104, the controller 12 calculates a predetermined target heat generation Q_t based on the temperature T_q of the discharged oxidant gas. Accordingly, in the temperature control of the fuel cell 20, the controller 12 sets the target heat generation Q_t such that the temperature T_q of the discharged oxidant gas does not exceed the heatproof limit temperature T_k of the internal reforming catalyst 22. That is, the controller 12 is configured to act as a target heat generation calculation unit and/or a target heat generation setting unit. The target heat generation Q_t is a target value of a heat generation of partial oxidation reforming. The target heat generation Q_t is a target value of a heat generation per predetermined time (for example, a detection cycle of the temperature T_q of the discharged oxidant gas). For example, the target heat generation Q_t can be obtained by a product of the temperature increase limit ΔT_m , a specific heat Sh of the cell stack 21 or the cells, and a mass Mc of the cells in which partial oxidation reforming is occurred, that is, $Q_t = \Delta T_m \times Sh \times Mc$. The controller 12 calculates, based on the heat generation obtained from a reaction formula of partial oxidation reforming, a flow rate of a fuel (and oxygen) for achieving the target heat generation Q_t .

(42) In the fuel flow rate calculation step S105, the controller 12 calculates a fuel flow rate F_f , which is a flow rate of the raw fuel, based on the target heat generation Q_t . That is, the controller 12 is configured to act as a fuel flow rate calculation unit. The fuel flow rate F_f is a flow rate of the raw fuel necessary for achieving the target heat generation Q_t by partial oxidation reforming. When power generation is started in the cell stack 21, the fuel flow rate F_f calculated in the fuel flow rate calculation step S105 is a flow rate of the raw fuel for maintaining the power generation while achieving the target heat generation Q_t .

(43) In the mixed air flow rate calculation step S106, the controller 12 calculates a mixed air flow rate F_a , which is an amount of air to be mixed with the raw fuel, based on the target heat generation Q_t . That is, the controller 12 is configured to act as a mixed air flow rate calculation unit. The mixed air flow rate F_a is a flow rate of the air necessary to be mixed with the raw fuel in order to achieve the target heat generation Q_t by partial oxidation reforming. When the power generation is started in the cell stack 21, the mixed air flow rate F_a is a flow rate of air containing oxygen

corresponding to an amount of partial oxidation reforming to be occurred.

(44) In the control execution step **S107**, the controller **12** controls the fuel flow rate adjustment device **31** and the air flow rate adjustment device **33** so as to achieve the calculated fuel flow rate F_f and mixed air flow rate F_a . That is, the controller **12** controls the fuel flow rate F_f and the mixed air flow rate F_a such that the heat generation of partial oxidation reforming in the cell stack **21** is the target heat generation Q_t . Therefore, the controller **12** is configured to act as a fuel flow rate control unit that controls, by using the fuel flow rate adjustment device **31**, the flow rate of the raw fuel to be supplied to the fuel cell **20**. The controller **12** is also configured to act as a mixed air flow rate control unit that controls, by using the air flow rate adjustment device **33**, the flow rate of the air to be mixed with the raw fuel. In this way, by controlling the fuel flow rate F_f and the mixed air flow rate F_a , the temperature of the fuel cell **20** is controlled within a range not exceeding the heatproof limit temperature T_k of the internal reforming catalyst **22** even when a temperature distribution that decreases from the anode inlet **A1** to the anode outlet **A2** is generated inside the fuel cell **20** by partial oxidation reforming. Therefore, even when the fuel is reformed by partial oxidation reforming inside the cell stack **21**, the fuel cell system **10** can continuously and efficiently operate without damaging the internal reforming catalyst **22**.

(45) As described above, the temperature control of the fuel cell **20** performed with the heat generated by partial oxidation reforming in the cell stack **21** is particularly useful in an operation scene in which the warm-up operation is performed at the time of starting the fuel cell system **10** and in an operation scene in which the temperature of the cell stack **21** is recovered when the temperature of the cell stack **21** decreases due to steam reforming or the like. Hereinafter, these operation scenes will be described in detail.

(46) FIG. 5 is a graph schematically showing a temperature control in the warm-up operation. A temperature T_a is the temperature at which partial oxidation reforming is possible, and is, for example, about 300°C . A temperature T_b is a temperature at which power generation is possible when the fuel cell **20** is a solid oxide fuel cell, and is, for example, about 500°C . A temperature T_c is a temperature that is maintained during a steady operation in consideration of power generation efficiency when the fuel cell **20** is a solid oxide fuel cell, and is, for example, about 800°C to about 1000°C . Graphs **G1** to **G6** in FIG. 5 represent respective temperatures of the oxidant gas measured at predetermined time intervals.

(47) When the fuel cell system **10** is started and the fuel cell **20** is to be warmed up to a temperature range in which the power generation is possible, the fuel cell **20** has a temperature lower than the temperature at which the power generation is possible. Therefore, the controller **12** supplies the oxidant gas and the raw fuel to the combustor **51** and thereby warm up the fuel cell **20** by flowing the heated oxidant gas. Accordingly, as shown in the graph **G1** and the graph **G2**, the fuel cell **20** is gradually warmed and approaches the temperature at which the power generation is possible. A temperature T_{d1} is the temperature T_q of the discharged oxidant gas at a time of the graph **G1**. A temperature T_{d2} is the temperature T_q of the discharged oxidant gas at a time of the graph **G2**.

(48) Here, when the above warm-up operation using heated oxidant gas is continued, as shown by intervals of the graphs **G1** to **G3**, the fuel cell **20** warms up slowly, and it takes a long time for the fuel cell **20** to reach the temperature at which the power generation is possible. Therefore, as shown in the graph **G3**, when the temperature T_q of the discharged oxidant gas reaches a temperature T_{d3} and is equal to or higher than the temperature T_a at which partial oxidation reforming is possible, the controller **12** calculates a temperature increase limit **63** which is the temperature increase limit ΔT_m at a time of the graph **G3** by subtracting the detected temperature T_{d3} of the discharged oxidant from the heatproof limit temperature T_k . The controller **12** calculates the target heat generation Q_t based on the temperature increase limit **63**, and further calculates the fuel flow rate F_f and the mixed air flow rate F_a based on the calculated target heat generation Q_t . Thereafter, when the controller **12** controls the fuel flow rate adjustment device **31** and the air flow rate adjustment device **33** according to the calculated fuel flow rate F_f and mixed air flow rate F_a ,

partial oxidation reforming proceeds in a predetermined range R_{pox} of the cell stack **21** as shown in a graph $G3^*$. As a result, a temperature of the predetermined range R_{pox} increases with the heat generated by partial oxidation reforming. However, even in the predetermined range R_{pox} , a maximum temperature of the fuel cell **20** is equal to or lower than the heatproof limit temperature T_k .

(49) The heat generated by partial oxidation reforming in the graph $G3^*$ propagates through the cell stack **21**. Therefore, at a next timing of detecting the temperature T_q of the discharged oxidant gas as shown in the graph $G4$, an interval between the graph $G3$ and the graph $G4$ is larger than the intervals between the graphs $G1$ to $G3$. That is, in the operation scene in which the warm-up operation is performed, the warm-up of the fuel cell **20** is promoted by occurring partial oxidation reforming.

(50) At the time of the graph $G4$, the temperature T_q of the discharged oxidant gas is a temperature T_{d4} , which is lower than the temperature T_b at which the power generation is possible, but higher than the temperature T_a at which partial oxidation reforming is possible. Therefore, as described above, the controller **12** gives rise to partial oxidation reforming again in the cell stack **21**, and promotes the warm-up of the fuel cell **20** with the heat of partial oxidation reforming. That is, the controller **12** calculates a temperature increase limit **64** at the time of the graph $G4$. Then, the controller **12** calculates the target heat generation Q_t based on the temperature increase limit **64**, and further calculates and controls the fuel flow rate F_f and the mixed air flow rate F_a based on the calculated target heat generation Q_t . Accordingly, as shown in a graph $G4^*$, partial oxidation reforming is occurred in the predetermined range R_{pox} of the cell stack **21**, and the heat is propagated to promote the warm-up of the fuel cell **20** again.

(51) Since the temperature increase limit **64** at the time of the graph $G4$ is smaller than the temperature increase limit **63** at the time of the graph $G3$, the target heat generation Q_t at the time of the graph $G4$ is naturally smaller than the target heat generation Q_t at the time of the graph $G3$. The fuel flow rate F_f and the mixed air flow rate F_a at the time of the graph $G4$ are also smaller than the fuel flow rate F_f and the mixed air flow rate F_a at the time of the graph $G3$. As a result, a heat generation of partial oxidation reforming occurred in the predetermined range R_{pox} of the cell stack **21** is reduced at the time of the graph $G4$ than at the time of the graph $G3$. As a result, even when partial oxidation reforming is additionally occurred at the time of the graph $G4$, the maximum temperature of the fuel cell **20** is maintained at a value equal to or lower than the heatproof limit temperature T_k .

(52) Thereafter, as shown in the graph $G5$, when the temperature T_q of the discharged oxidant gas becomes a temperature T_{d5} and exceeds the temperature T_b at which the power generation is possible, the power generation is started in the cell stack **21** by the supply of the fuel and the oxidant gas. Generally, the fuel cell **20** has higher power generation efficiency as the temperature increases. Therefore, the fuel cell **20** is operated while maintaining a high temperature as much as possible within a range equal to or lower than the heatproof limit temperature T_k . Therefore, even after the power generation is started, as shown by the graph $G5$, when the target temperature T_c of the steady operation is not achieved, the controller **12** continues to warm up the fuel cell **20** in the same manner as described above. That is, the controller **12** calculates a temperature increase limit **65** at a time of the graph $G5$. Then, the controller **12** calculates the target heat generation Q_t based on the temperature increase limit **65**, and further calculates and controls the fuel flow rate F_f and the mixed air flow rate F_a based on the calculated target heat generation Q_t . Accordingly, as shown in the graph $G4^*$, partial oxidation reforming is generated in the predetermined range R_{pox} of the cell stack **21**, and the heat is propagated to promote the warm-up of the fuel cell **20**. As a result, at a time of the graph $G6$, the temperature T_q of the discharged oxidant gas is a temperature T_{d6} , and the temperature of the entire fuel cell **20** also exceeds the temperature T_c to be maintained during the steady operation. Accordingly, as compared with a case where the warm-up operation is performed by flowing the oxidant gas heated only by a heater or the combustor **51** as in related art,

the warm-up operation is completed in a short time and the fuel cell system **10** can shift to the steady operation in which the power generation is performed in the cell stack **21**.

(53) FIG. **6** is a graph schematically showing a temperature control in steady operation. Even when the temperature of the fuel cell **20** generally exceeds the temperature T_c that is should to be maintained during the steady operation and the steady operation is continued, the temperature of the fuel cell **20** may decrease due to endothermic of steam reforming or the like, a load on the fuel cell **20**, or the like. For example, when an output power required for the fuel cell **20** is at a middle or low level, and the steady operation is performed with a low load or a middle load on the fuel cell **20**, the temperature distribution in the fuel cell **20** becomes a temperature distribution in which the temperature generally increases from the cathode inlet **C3** to the cathode outlet **C4**. This is because, when the fuel cell **20** of the present embodiment is operated at low or middle load, the heat generated by the power generation is small, but endothermic of steam reforming or the like is even smaller, and thus the heat generated by the power generation becomes higher than endothermic of steam reforming or the like. On the other hand, when an output power required for the fuel cell **20** is large and a load on the fuel cell **20** is high, the temperature distribution in the fuel cell **20** may become a temperature distribution in which the temperature decreases from the cathode inlet **C3** to the cathode outlet **C4**. This is because, when the fuel cell **20** of the present embodiment is operated at high load, the heat generated by the power generation is large, but endothermic of steam reforming or the like is further large, and thus the heat generated by the power generation becomes lower than endothermic of steam reforming or the like. In this way, when the steady operation is continued in a state where the fuel cell **20** has a high load, for example, as shown in a graph **G7**, the temperature of the fuel cell **20** may become lower than the temperature T_c to be maintained during the steady operation even after the start of the steady operation. In such a state, when the steady operation is simply continued, the temperature may be subsequently lower than the temperature T_b at which the power generation is possible, and the power generation may not be continued.

(54) Therefore, the controller **12** continues to detect the temperature T_q of the discharged oxidant gas even after shifting to the steady operation. Then, when the temperature T_q of the discharged oxidant gas is equal to or lower than the temperature T_c to be maintained during the steady operation, the controller **12** heats the fuel cell **20** with the heat generated by partial oxidation reforming. For example, as shown in the graph **G7**, when the temperature T_q of the discharged oxidant gas becomes a temperature T_{d7} and becomes lower than the temperature T_c to be maintained during the steady operation, the controller **12** calculates a temperature increase limit **67**. Then, the controller **12** calculates the target heat generation Q_t based on the temperature increase limit **67**, and further calculates and controls the fuel flow rate F_f and the mixed air flow rate F_a based on the calculated target heat generation Q_t . Accordingly, as shown in a graph **G7***, partial oxidation reforming is occurred in the predetermined range R_{pox} of the cell stack **21**. As the heat generated by partial oxidation reforming propagates, the temperature of the fuel cell **20** is equal to or higher than the temperature T_c to be maintained during the steady operation, as shown in a graph **G8**. As a result of the temperature control, the fuel cell **20** can continue the power generation while satisfying the required output power even when the high load state continues. Also, in a case of performing the temperature control for maintaining the temperature of the fuel cell **20** at a temperature equal to or higher than the temperature T_c to be maintained during the steady operation as described above, the controller **12** performs the temperature control based on the temperature T_q of the discharged oxidant gas, and thus the temperature of the fuel cell **20** does not exceed the heatproof limit temperature T_k even when observing locally. Therefore, the fuel cell system **10** can stably continue the steady operation without damaging the fuel cell **20** or the like.

(55) As described above, a control method for the fuel cell system according to the embodiment is a control method for the fuel cell system **10** includes the fuel cell **20** including the cell stack **21** that has the internal reforming catalyst **22** for generating hydrogen from hydrocarbon, the first flow path **23** that supplies a fuel containing hydrocarbon to the cell stack **21**, and the second flow path

24 that supplies the oxidant gas to the cell stack **21** such that the oxidant gas flows oppositely or orthogonally to the fuel. Wherein, the control method includes detecting the temperature T_q of the discharged oxidant gas that is the oxidant gas discharged from the second flow path **24** and performing the temperature control of the fuel cell **20** based on the temperature T_q of the discharged oxidant gas.

(56) In the control method for the fuel cell system **10**, as described above, the fuel cell **20** is of the internal reforming type, and thus the temperature may be relatively higher in the vicinity of the anode inlet **A1** than in the central portion of the cell stack **21**. The fuel cell **20** forms a so-called counter flow or cross flow, and the temperature of the cathode outlet **C4** may be regarded as the temperature of the anode inlet **A1** which is a relatively high temperature portion. The controller **12** of the fuel cell system **10** performs the temperature control of the fuel cell **20** based on the temperature T_q of the discharged oxidant gas detected at the cathode outlet **C4** close to the anode inlet **A1**. That is, in the control method for the fuel cell system **10** according to the above embodiment, the temperature control is performed based on the temperature of the relatively high temperature portion in the fuel cell **20**, and thus it is possible to realize a stable control more reliably while observing the heatproof limit temperature T_k which is a heat resistance standard.

(57) On the other hand, in a case where the temperature control of the fuel cell **20** is performed based on the temperature of the central portion of the fuel cell **20** as in the related art, in an operation scene in which the fuel cell **20** is of the internal reforming type and the temperature is relatively higher in the vicinity of the anode inlet **A1** than in the central portion of the cell stack **21**, the relatively high temperature portion in the vicinity of the anode inlet **A1** may have a temperature exceeding the heatproof limit temperature T_k which is the heat resistance standard, and the fuel cell **20** may be damaged or the like. Therefore, by the control method for the fuel cell system **10** according to the above embodiment, the fuel cell **20** can be operated more reliably and stably without being damaged or the like, as compared with a related-art control method for a fuel cell system.

(58) As an example of a scene in which the vicinity of the anode inlet **A1** is more likely to be the relatively high temperature portion than the central portion of the cell stack **21**, there is an operation scene in which partial oxidation reforming is generated. Therefore, the above control method for the fuel cell system **10** is particularly suitable when partial oxidation reforming is occurred in the cell stack **21**.

(59) For example, when partial oxidation reforming is actively occurred from a low temperature range of about 300° C. in order to promote the warm-up (temperature increase) of the fuel cell **20**, the anode inlet **A1** side reaches the maximum temperature in the temperature distribution in the fuel cell **20** due to the heat generated by partial oxidation reforming. Therefore, even when partial oxidation reforming is occurred for warming up the fuel cell **20**, by detecting the temperature T_q of the discharged oxidant gas with the temperature sensor **60** provided at the cathode outlet **C4** and performing the temperature control of the fuel cell **20** based on the detected temperature T_q of the discharged oxidant gas as described above, it is possible to promote the warm-up of the fuel cell **20** within a range not exceeding the heatproof limit temperature T_k even when observing locally. In a case where the fuel and the oxidant gas form the counter flow as in the above embodiment, the temperature T_q of the discharged oxidant gas detected by the temperature sensor **60** is substantially equal to the maximum temperature of the fuel cell **20**, and thus the above temperature control can be performed more accurately.

(60) In the control method for the fuel cell system **10** according to the above embodiment, the temperature control of the fuel cell **20** is performed particularly such that the temperature of the fuel cell **20** is equal to or lower than the heatproof limit temperature T_k , which is a predetermined temperature. The heatproof limit temperature T_k is optional. Therefore, the control method for the fuel cell system **10** according to the above embodiment is particularly suitable when the temperature of the fuel cell **20** is controlled to be equal to or lower than the predetermined

temperature, and it is possible to realize a stable control in which a risk of damage or the like of the fuel cell **20** and the fuel cell system **10** is reduced.

(61) In addition to the above basic configuration, the fuel cell system **10** according to the above embodiment further includes the fuel flow rate adjustment device **31** that adjusts the flow rate of the fuel to be supplied to the fuel cell **20**, and the air flow rate adjustment device **33** that adjusts the flow rate of the air to be mixed with the fuel. In the control method for the fuel cell system **10** according to the above embodiment, specifically, it controls the fuel flow rate F_f which is the flow rate of the fuel and the mixed air flow rate F_a which is the flow rate of the air to be mixed with the fuel such that the heat generation by partial oxidation reforming in the cell stack **21** becomes the predetermined target heat generation Q_t based on the temperature T_q of the discharged oxidant gas. Therefore, the control method for the fuel cell system **10** according to the above embodiment realizes the temperature control of the fuel cell **20** in a manner of using partial oxidation reforming that is inevitably generated due to properties of the internal reforming catalyst **22**. That is, in the control method for the fuel cell system **10** according to the above embodiment, the temperature control of the fuel cell **20** is realized without requiring preparation of a special heat source or cold source.

(62) In the fuel cell system **10**, partial oxidation reforming is actively occurred by mixing the air with the fuel using the air flow rate adjustment device **33**. In the fuel cell system **10**, the target heat generation Q_t is determined based on the temperature T_q of the discharged oxidant gas, and the fuel flow rate F_f and the mixed air flow rate F_a are controlled. As a result, in the fuel cell system **10**, the temperature of the fuel cell **20** is controlled by controlling a heat generation of partial oxidation reforming to be occurred. In this way, when the heat generated by partial oxidation reforming is controlled based on the temperature T_q of the discharged oxidant gas, the temperature control of the fuel cell **20** can be suitably performed. Particularly, in a case of warming up the fuel cell **20** and in a case of maintaining the temperature of the fuel cell **20** in order to continuously use the fuel cell **20** at a high load, the fuel cell system **10** can particularly appropriately control the temperature of the fuel cell **20** within the range not exceeding the heatproof limit temperature T_k even when observing locally. Regarding the warm-up of the fuel cell **20**, the warm-up can be promoted by the above temperature control using the heat generated by partial oxidation reforming. When the fuel cell **20** is used at a high load, the temperature T_c to be maintained for the steady operation can be maintained.

(63) In the control method for the fuel cell system **10** according to the above embodiment, specifically, the target heat generation Q_t is set such that the temperature T_q of the discharged oxidant gas does not exceed the heatproof limit temperature T_k of the internal reforming catalyst **22** which is the reforming catalyst. Accordingly, the temperature of the fuel cell **20** can be more appropriately controlled within the range not exceeding the heatproof limit temperature T_k even when observing locally.

(64) In the control method for the fuel cell system **10** according to the above embodiment, specifically, when the temperature T_q of the discharged oxidant gas is equal to or higher than the temperature at which partial oxidation reforming is possible in the cell stack **21**, the temperature control of the fuel cell **20** is performed, for example, by supplying the fuel and the air to be mixed with the fuel. Accordingly, the temperature control of the fuel cell **20** using the heat generated by partial oxidation reforming can be performed by properly supplying a mixture of the fuel and the air only when the temperature reaches a temperature range in which partial oxidation reforming is possible. As a result, it is possible to eliminate waste of the fuel and to prevent failures such as precipitation of carbon.

(65) The fuel cell system **10** according to the above embodiment includes: the cell stack **21** including the internal reforming catalyst **22** which is the reforming catalyst for generating hydrogen from hydrocarbon; the first flow path **23** that supplies the fuel containing hydrocarbon to the cell stack **21**; the second flow path **24** that supplies the oxidant gas to the cell stack **21** such that the

oxidant gas flows oppositely or orthogonally to the fuel; the temperature sensor **60** that detects the temperature T_q of the discharged oxidant gas that is the oxidant gas discharged from the second flow path **24**; and the controller **12** that performs the temperature control of the fuel cell **20** based on the temperature T_q of the discharged oxidant gas.

(66) The configuration of the fuel cell system **10** can be changed without departing from the gist of the temperature control of the fuel cell **20** according to the above embodiment. For example, in the above embodiment, an example in which the temperature control of the fuel cell **20** is performed using the heat generated by partial oxidation reforming has been described, but the temperature of the fuel cell **20** may be controlled by a method other than using the heat generated by partial oxidation reforming, such as heating by a heater, when the temperature control of the fuel cell **20** can be performed based on the temperature T_q of the discharged oxidant gas. For example, instead of using the heat generated by partial oxidation reforming, the controller **12** may control an amount of air supplied by the oxidant gas supply device **41** or an amount of a fuel, a discharged gas, and/or new air supplied to the combustor **51** such that the temperature T_q of the discharged oxidant gas is a desired temperature (for example, to comply with a predetermined upper limit temperature). However, as in the above embodiment, the temperature control of the fuel cell **20** using partial oxidation reforming is particularly preferred because a special configuration for controlling the temperature of the fuel cell **20** is not necessary.

Second Embodiment

(67) The fuel cell system **10** according to the above first embodiment includes one internal reforming fuel cell **20** including the cell stack **21** including the internal reforming catalyst **22**. Such an internal reforming type fuel cell **20** is used with a somewhat excessive supply of the raw fuel in order to prevent precipitation of carbon or the like. Therefore, a utilization rate of the fuel is low. Therefore, the fuel cell system **10** according to the above first embodiment can improve a fuel utilization rate by further using another fuel cell in combination as follows.

(68) FIG. 7 is a block diagram showing a configuration of a fuel cell system **210** according to a second embodiment. The fuel cell system **210** according to the second embodiment is obtained by using the fuel cell **20** of the fuel cell system **10** according to the first embodiment as a first fuel cell and adding a second fuel cell **220**. Therefore, configurations common to those of the fuel cell system **10** according to the first embodiment are designated by the same reference numerals as those in the first embodiment, and description thereof will be omitted.

(69) The second fuel cell **220** is a fuel cell including a cell stack **221** including a smaller amount of the internal reforming catalyst **22** as compared with the fuel cell **20** as the first fuel cell. In the present embodiment, the cell stack **221** of the second fuel cell **220** is a solid oxide fuel cell that does not include the internal reforming catalyst **22**. That is, the second fuel cell **220** generates a power by being supplied with a reformed fuel.

(70) An anode inlet **A3** of the second fuel cell **220** is connected to the anode outlet **A2** of the fuel cell **20**. An anode outlet **A4** of the second fuel cell **220** is connected to the combustor **51** via the fuel discharge path **34**. That is, a fuel discharged from the fuel cell **20** is supplied to the second fuel cell **220** as a fuel. The raw fuel is supplied with a somewhat excessive, and thus the discharged fuel of the fuel cell **20** is not completely used as the fuel, hydrogen remains, and the discharged fuel can still be used as a fuel. On the other hand, for the discharged fuel of the fuel cell **20**, almost all of the raw fuel is reformed into hydrogen by the internal reforming of the fuel cell **20**. Therefore, the discharged fuel of the fuel cell **20** is used as the reformed fuel in the second fuel cell **220**.

(71) A cathode inlet **C1** of the second fuel cell **220** is connected to the oxidant gas supply device **41** via the oxidant gas supply path **42**. A cathode outlet **C2** of the second fuel cell **220** is connected to the cathode inlet **C3** of the fuel cell **20**. That is, the second fuel cell **220** uses an oxidant gas supplied from the oxidant gas supply device **41** for power generation. Thereafter, the second fuel cell **220** supplies an oxidant gas discharged from the second fuel cell **220** (a discharged oxidant gas) to the fuel cell **20** as an oxidant gas. The discharged oxidant gas of the second fuel cell **220**

contains oxygen, and can also sufficiently act as the oxidant gas in the fuel cell **20**.

(72) As described above, the fuel cell system **210** according to the second embodiment includes the second fuel cell **220** in addition to the fuel cell **20** which is the first fuel cell. The fuel discharged from the fuel cell **20** is supplied to the second fuel cell **220**. As a result, in the internal reforming fuel cell **20**, the raw fuel is supplied and used with a somewhat excessive, and thus the utilization rate of the fuel is low, but the discharged fuel of the fuel cell **20** is reused for the power generation in the second fuel cell **220**. Therefore, the fuel cell system **210** can use the raw fuel for the power generation with almost no remain, and can improve the utilization rate of the fuel as compared with the fuel cell system **10** according to the first embodiment.

(73) The fuel cell system **210** according to the second embodiment supplies the discharged oxidant gas, which is the oxidant gas discharged from the second fuel cell **220**, to the fuel cell **20** as the oxidant gas. When an endothermic reaction such as steam reforming proceeds inside the fuel cell **20**, the temperature of the fuel cell **20** decreases, but as described above, the discharged oxidant gas heated by the power generation in the second fuel cell **220** is supplied to the fuel cell **20** as the oxidant gas, and thus a heat balance between the heat generated by the power generation in the fuel cell **20** and endothermic by fuel reforming can be easily maintained. The fuel cell system **210** as a whole can also easily maintain a heat balance. Therefore, the controller **12** can easily maintain the internal reforming fuel cell **20** in a steady operation more stably and with high efficiency.

(74) For the second fuel cell **220** added in the fuel cell system **210** according to the second embodiment, one with less the internal reforming catalyst **22** than the fuel cell **20** is adopted. Therefore, in the second fuel cell **220**, there is less occurrence of endothermic reactions such as steam reforming. Therefore, by supplying the discharged oxidant gas of the second fuel cell **220** to the fuel cell **20**, the controller **12** can particularly easily maintain the heat balance between the heat generated by the power generation in the fuel cell **20** and endothermic by the fuel reforming. The fuel cell system **210** as a whole can also easily maintain the heat balance. When the second fuel cell **220** does not include the internal reforming catalyst **22** as in the second embodiment above, it is easiest to maintain the heat balance.

Modification

(75) In the first embodiment and the second embodiment described above, the temperature sensor **60** is provided in the cathode outlet **C4** at which a plurality of second flow paths **24** branched in the cell stack **21** are integrated again into one, but arrangement of the temperature sensor **60** can be changed optionally within a range in which the maximum temperature in the temperature distribution in the fuel cell **20** can be detected or estimated. For example, FIG. **8** is an explanatory diagram of a modification in which an attachment position of the temperature sensor **60** is changed. As shown in FIG. **8**, a flow path **301** is provided in which a part of the second flow path **24** branched in the cell stack **21** is drawn out and merges into the cathode outlet **C4**, and the temperature sensor **60** can be provided in the flow path **301**. In this way, when the temperature sensor **60** is provided in the flow path **301** in which a part of the second flow path **24** is drawn out, even when a temperature distribution is formed in the fuel cell **20** in a Y direction perpendicular to the first flow path **23** and the second flow path **24** due to convenience of an overall structure of the fuel cell **20**, the controller **12** can easily accurately detect or estimate the maximum temperature of the fuel cell **20** regardless of the temperature distribution in the Y direction.

(76) In the first embodiment and the second embodiment described above, the fuel and the oxidant gas in the fuel cell **20** form the counter flow, but the fuel and the oxidant gas in the fuel cell **20** may form a so-called cross flow. For example, FIG. **9** is an explanatory diagram of a modification in which a flow path configuration of the fuel and the oxidant gas is changed. That is, instead of the counter-flow cell stack **21**, the fuel cell **20** may adopt a cell stack **302** in which the second flow path **24** is a flow path for supplying the oxidant gas to the cell stack **21** such that the oxidant gas flows orthogonally to the fuel. In this way, also in a case where the cell stack **302** in which the fuel and the oxidant gas form the so-called cross flow is adopted, the control methods according to the

first embodiment and the second embodiment described above can be performed, and effects thereof can be obtained. This is because, also in the cross-flow cell stack **302**, the temperature of the discharged oxidant gas has a constant correlation with the maximum temperature in the temperature distribution in the fuel cell **20**, and thus the controller **12** can estimate the maximum temperature of the fuel cell **20** based on the temperature of the discharged oxidant gas.

(77) A parallel-flow cell stack may also be adopted in which the first flow path **23** and the second flow path **24** are made parallel to each other, and the fuel and the oxidant gas are made to flow through these flow paths in parallel to each other. Also in this case, the control methods according to the first embodiment and the second embodiment described above can be performed, and the effects thereof can be obtained. However, in order to make the temperature distribution in the fuel cell **20** uniform and increase the power generation efficiency regardless of whether partial oxidation reforming is occurred, it is preferable to adopt the counter-flow cell stack **21** or the cross-flow cell stack **302**, and it is particularly preferable to adopt the counter-flow cell stack **21**.

(78) FIG. **10** is an explanatory diagram of a modification in which the flow path configuration of the fuel and the oxidant gas and the attachment position of the temperature sensor **60** are changed with respect to the first embodiment and the second embodiment described above. In the modification of the fuel cell **20** in FIG. **10**, the cross-flow cell stack **302** is adopted, and a flow path **303** is provided in which a part of the second flow path **24** is drawn out and merges into the cathode outlet **C4**. The temperature sensor **60** is provided in the flow path **303**. As described above, in a case where the cross-flow cell stack **302** is adopted, when the flow path **303** is provided in which a part of the second flow path **24** is drawn out and the temperature sensor **60** is provided in the flow path **303**, a correlation between the maximum temperature in the temperature distribution in the fuel cell **20** and the temperature T_q of the discharged oxidant gas to be detected is high. Therefore, when the controller **12** estimates the maximum temperature of the fuel cell **20**, an accuracy is increased, and as a result, an accuracy of the temperature control of the fuel cell **20** is increased. Particularly, as shown in FIG. **10**, when the flow path **303** is provided on the anode inlet **A1** side where a relatively large amount of partial oxidation reforming is occurred, in a case where a portion having a maximum temperature is generated on the anode inlet **A1** side in the fuel cell **20** due to the heat generated by partial oxidation reforming, a temperature of the portion can be detected directly and accurately. Therefore, the temperature control of the fuel cell **20** by the controller **12** can be performed more accurately.

(79) Although the embodiments of the present invention have been described above, the above embodiments are merely a part of application examples of the present invention, and are not intended to limit the technical scope of the present invention to the specific configurations of the above embodiments. Various changes and modifications may be made to the above embodiments within the scope of the matters described in the claims. The technical ideas described in the embodiments and the modifications described above may be appropriately combined.

Claims

1. A control method for a fuel cell system that comprises a fuel cell comprising a cell stack, the cell stack comprising a reforming catalyst for generating hydrogen from hydrocarbon, a first flow path configured to supply a fuel containing hydrocarbon to the cell stack, a second flow path configured to supply an oxidant gas to the cell stack such that the oxidant gas flows oppositely or orthogonally to the fuel, a fuel flow rate adjustment device configured to adjust a flow rate of the fuel to be supplied to the fuel cell, and an air flow rate adjustment device configured to adjust a flow rate of air to be mixed with the fuel, the control method comprising: detecting, in an oxidant gas discharge path, a temperature of a discharged oxidant gas, which is the oxidant gas discharged from the second flow path; performing a temperature control of the fuel cell based on the temperature of the discharged oxidant gas when partial oxidation reforming occurs in the cell stack by supplying air to

- the first flow path; and controlling the flow rate of the fuel and the flow rate of the air such that a heat generation by partial oxidation reforming in the cell stack becomes a predetermined target heat generation based on the temperature of the discharged oxidant gas, wherein the target heat generation is set such that the temperature of the discharged oxidant gas does not exceed a heatproof limit temperature of the reforming catalyst in the temperature control of the fuel cell.
2. The control method for a fuel cell system according to claim 1, wherein: the temperature control of the fuel cell is performed when the temperature of the discharged oxidant gas is equal to or higher than a temperature at which partial oxidation reforming is possible in the cell stack.
 3. A fuel cell system, comprising: a first fuel cell comprising a cell stack comprising a reforming catalyst for generating hydrogen from hydrocarbon, a first flow path configured to supply a fuel containing hydrocarbon to the cell stack, and a second flow path configured to supply an oxidant gas to the cell stack such that the oxidant gas flows oppositely or orthogonally to the fuel; a temperature sensor located in an oxidant gas discharge path and configured to detect a temperature of a discharged oxidant gas, which is the oxidant gas discharged from the second flow path; a fuel flow rate adjustment device configured to adjust a flow rate of the fuel to be supplied to the first fuel cell; and an air flow rate adjustment device configured to adjust a flow rate of air to be mixed with the fuel; and a controller configured to: perform a temperature control of the first fuel cell based on the temperature of the discharged oxidant gas when partial oxidation reforming occurs in the cell stack by supplying air to the first flow path, and control the flow rate of the fuel and the flow rate of the air such that a heat generation by partial oxidation reforming in the cell stack becomes a predetermined target heat generation based on the temperature of the discharged oxidant gas, wherein the target heat generation is set such that the temperature of the discharged oxidant gas does not exceed a heatproof limit temperature of the reforming catalyst in the temperature control of the first fuel cell.
 4. The fuel cell system according to claim 3, further comprising: a second fuel cell to which the fuel discharged from the first fuel cell is supplied.
 5. The fuel cell system according to claim 4, wherein: the first fuel cell is configured to receive an oxidant gas discharged from the second fuel cell.
 6. The fuel cell system according to claim 4, wherein: the second fuel cell has less reforming catalyst than the first fuel cell.
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